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PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

Music Playlist Manager

Team Name: *Team #*	*Dictators* *DSCPP-III-2026-T326*
Team member 1 (Team Lead) *(Last Name, name: student ID, email, picture):*	*Rawat,Tanishq – 2510012727* *tanishqrawat201@gmail.com* 
Team member 2 *(Last Name, name: student ID, email, picture):*	*Priyanshu, Lucky – 2510370435* *luckypriyanshu879@gmail.com* 

PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

(Brief restatement of your project's main goal. Max 300 words).

Music Playlist Manager is a C++ based project developed using Object-Oriented Programming (OOP), Data Structures and Algorithms (DSA), and File Handling.

The main objective of the project is to manage songs and playlists in a simple and organized way. The system provides a Preloaded Songs Library, song searching and filtering, playlist management, queue management, recently played songs, favorites, shuffle, repeat, play count and smart recommendations.

The project uses a Doubly Linked List for playlist management and next/previous navigation, Queue for upcoming songs, Stack for recently played songs, and searching and sorting techniques for library operations and listening statistics.
The project also demonstrates OOP concepts through classes such as Song, MusicLibrary, Playlist, QueueManager, MusicPlayer, RecommendationSystem, User, NormalUser, PremiumUser and FileManager.

Updated Project Approach and Architecture (2 pts)

(Describe your current approach, including system design, communication protocols, libraries used, etc. Max 300 words).

Our project follows a modular and class based C++ approach every function of the music playlist manager is handled using separate classes and data structures

The main classes are song,music library,Playlist, queue manager, music player, recommendation system, user, normal user, premium user and file manager here doubly linked list is used for playlist management and next/previous navigation. A Queue is used to manage upcoming songs, while a Stack is used for recently played songs. Searching techniques are used for finding songs by title, artist, genre and mood. Sorting techniques are used for play count based operations such as Most Played and Least Played songs.

Also the recommadation sysytem here uses song information such as genre, mood, listening history and play count to provide accurate recommendations.
The user class represents inheritance through NormalUser and PremiumUser classes. File Handling is used to save and load required project data so the information can be maintained between program runs.

Tasks Completed (7 pts)

(Describe the main tasks that have been assigned and already completed. Max 250 words).

Task Completed	Team Member
Song Class Song attributes Preloaded Songs Library Music Library Search and Filtering Playlist using doubly linked list Queue and Recently Played Stack	Tanishq Rawat
Favorites Shuffle and Repeat Play Count Most/Least Played Smart Recommendations NormalUser PremiumUser Classes File Handling	Lukky Priyanshu

Challenges/Roadblocks (7 pts)

(Describe the challenges that you have faced or are facing so far and how you plan to solve them. Max 300 words).

The main challenges faced during development were managing playlist navigation using pointers in the Doubly Linked List and maintaining correct next and previous operations.

Another challenge was selecting suitable data structures for different features. The Doubly Linked List was selected for playlists, Queue for upcoming songs and Stack for recently played songs.

Integrating different OOP classes was also a challenge because the classes need to work together without making the program difficult to manage.

Although maintaining play counts and using sorting for most played and least played songs required careful handling of song data.
File Handling also required proper save and load operations so that the project data can be maintained correctly.

These challenges are being handled through modular class design, testing each feature separately and performing final integrated testing after combining all modules.

Tasks Pending (7 pts)

(Describe the main tasks that you still need to complete. Max 250 words).

Task Pending	Team Member (to complete the task)
Final Integrated Testing	Tanishq Rawat
Edge-Case Testing and Final Bug Fixing	Lukky Priyanshu
Final Presentation and Demonstration Preparation	Both

Project Outcome/Deliverables (2 pts)

(Describe what are the key outcomes / deliverables of the project. Max 200 words).

Our project provides a functional C++ based music playlist manager that presents the practical use of OOP and DSA

And the main deliverables include the Song Class,Preloaded Songs Library, Music library,Search and Filtering,Playlist using Doubly Linked List, Queue, recently Played Stack, favorites, shuffle, Repeat, play Count,Most Played and Least Played,smart Recommendations, NormalUser and PremiumUser classes and file handling.

The project represents how different data structures and OOP concepts can be altogether to create a structured music playlist management system.

Progress Overview (2 pts)

(Summarize how much of the project is done, what's behind schedule, what's ahead of schedule. Max 200 words.)

Major Phase-II development of the Music Playlist Manager has been completed.

The core modules including the Song Class, Preloaded Songs Library, Music Library, Search and Filtering, Playlist, Queue, Recently Played Stack, favorites, shuffle, repeat, play count, Most/Least Played, smart Recommendations, normal user, premium user and File Handling have been completed.

The project is currently in the final stage and The remaining work includes a final integrated testing, edge case testing, final bug fixing and presentation preparation.

The project is progressing according to the planned scope and remains with C++, OOP, DSA and File Handling.
No major module is behind schedule.

Codebase Information (2 pts)

(Repository link, branch, and information about important commits.)

https://github.com/tanishqrawat201-svg/pbl

Song and music library — Added song class and preloaded songs library with search/filtering.
Playlist and doubly linked list — Implemented playlist creation, adding/removing songs and next/previous navigation.
Queue and recently played — Added queue for upcoming songs and stack for recently played songs.
Favorites, shuffle and repeat — Implemented these playlist/player features.
Play count and statistics — Added play count, most played and least played functionality.
Smart Recommendations — Added recommendations based on genre, mood and listening data.
User Classes — Added user, normal user and premium user using inheritance and polymorphism.
File handling — Added saving and loading of project data.
Integration and testing — combined the modules and tested the major features.

Testing and Validation Status (2 pts)

(Provide information about any tests conducted)

Test Type	**Status (Pass/Fail)**	**Notes**
Song and library operations	Pass	Song details and library operations were tested.
Search and filtering	Pass	Search by title, artist, genre and mood was tested.
Playlist and doubly linked list	Pass	Add, remove and next/previous operations were tested.
Queue and recently played stack	Pass	Queue and stack operations were tested.
Favorites, shuffle and repeat	Pass	Feature operations were tested.
Play count and sorting	Pass	Most played and least played operations were tested.
File handling	Pass	Save and load operations were tested.
Final integrated testing	Pending	Final combined testing and bug fixing are remaining.

Deliverables Progress (2 pts)

(Summarize the current status of all key project deliverables mentioned earlier. Indicate whether each deliverable is completed, in progress, or pending.)

Completed:

Song Class
Preloaded Songs Library
Search and Filtering
Playlist and Doubly Linked List
Queue
Recently Played Stack
Favorites
Shuffle and Repeat
Play Count
Most/Least Played
Smart Recommendations
Normal User and Premium User
Classes
File Handling.

Pending:

Final Integrated Testing
Audio demo files
Edge-Case Testing
Final Bug Fixing and Presentation Preparation.